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Contents

Apache Tasks

Task 1

First, we need to go to the directory and create virtual host inside:

```
[root@rhel-9 ~ #] cd /etc/httpd/conf.d
[root@rhel-9 /etc/httpd/conf.d #] vim my-virtual-host.conf
```

Its content:

```
<VirtualHost *:80>
    ServerAdmin mr-ravan
    ServerName idtech.local
    DocumentRoot "/var/www/html"
</VirtualHost>
```

As you can see, its data (like html files) located in the same directory with the default one.

Then we need to add apache port to firewall. First, let’s check firewall status:

```
[root@rhel-9 /etc/httpd/conf.d #] firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160 lo
  sources:
  services: cockpit dhcpv6-client ntp ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

As you can see, apache port (80) has not added yet. Let’s add it:

```
[root@rhel-9 /etc/httpd/conf.d #] firewall-cmd --add-port=80/tcp #Temporary
success
[root@rhel-9 /etc/httpd/conf.d #] firewall-cmd --permanent --add-port=80/tcp #Permanent(after reload)
success
[root@rhel-9 /etc/httpd/conf.d #] firewall-cmd --reload #Not mandatory
success
[root@rhel-9 /etc/httpd/conf.d #] firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160 lo
  sources:
  services: cockpit dhcpv6-client ntp ssh
  ports: 80/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

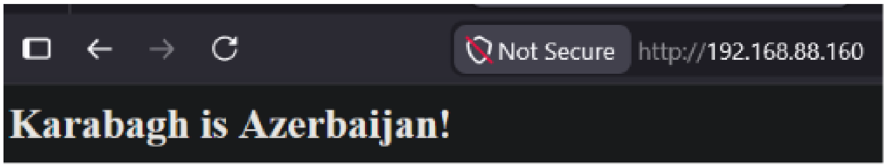
Then we restart the apache service:

```
[root@rhel-9 /etc/httpd/conf.d #] systemctl restart httpd
[root@rhel-9 /etc/httpd/conf.d #]
```

Now let’s redirect the html content to the index.html in default location:

```
[root@rhel-9 /etc/httpd/conf.d #] echo '<h2>Karabagh is Azerbaijan!</h2>' > /var/www/html/index.html
[root@rhel-9 /etc/httpd/conf.d #] cat /var/www/html/index.html
<h2>Karabagh is Azerbaijan!</h2>
```

Let’s check it from browser by typing “192.168.88.160:80”. (the first part is my ip and the second part is port of apache):



We can also use `curl` to check:

```
[root@rhel-9 ~ #] curl 192.168.88.160:80
<h2>Karabagh is Azerbaijan!</h2>
```

Task 2

We go to `/etc/httpd/conf.d/` directory to create and config virtual host. Let’s create a file:

```
[root@rhel-9 /etc/httpd/conf.d #] vim virtual-host-2.conf
```

Contents:

```
[root@rhel-9 /etc/httpd/conf.d #] cat virtual-host-2.conf
Listen 8282
<VirtualHost *:8282>
    ServerAdmin mr-ravan
    ServerName idtech.local
    DocumentRoot "/myapache"
</VirtualHost>
```

We create a directory and index.html:

```
[root@rhel-9 /etc/httpd/conf.d #] mkdir /myapache
[root@rhel-9 /etc/httpd/conf.d #] echo '<h2>RHCSA is my next step</h2>' > /myapache/index.html
[root@rhel-9 /etc/httpd/conf.d #] cat /myapache/index.html
<h2>RHCSA is my next step</h2>
```

Let’s add the port to firewall:

```
[root@rhel-9 ~ #] firewall-cmd --add-port=8282/tcp
success
[root@rhel-9 ~ #] firewall-cmd --permanent --add-port=8282/tcp
success
[root@rhel-9 ~ #] firewall-cmd --reload
success
[root@rhel-9 ~ #] firewall-cmd --list-all | grep -i "port"
  ports: 80/tcp 8282/tcp
```

Let’s add the port to SELinux:

```
[root@rhel-9 ~ #] semanage port -a -t http_port_t -p tcp 8282
```

Let’s add the new directory to SELinux:

```
[root@rhel-9 ~ #] semanage fcontext -a -t httpd_sys_content_t "/myapache(/.*)?"
[root@rhel-9 ~ #] restorecon -R -v /myapache
Relabeled /myapache from unconfined_u:object_r:default_t:s0 to unconfined_u:object_r:httpd_sys_content_t:s0
Relabeled /myapache/index.html from unconfined_u:object_r:default_t:s0 to unconfined_u:object_r:httpd_sys_content_t:s0
```

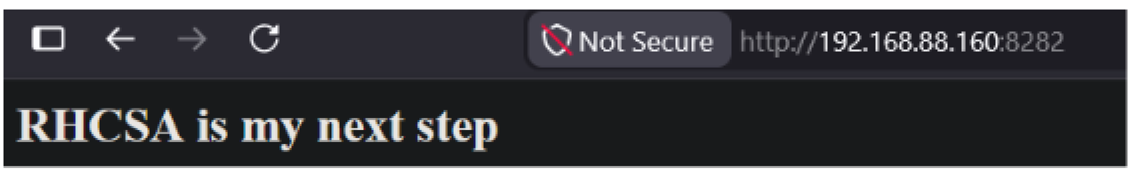
It is good to add (not mandatory) this to `/etc/httpd/conf/httpd.conf` file:

```
<Directory "/myapache">
    AllowOverride None
    # Allow open access:
    Require all granted
</Directory>
```

Then we need to restart httpd service:

```
[root@rhel-9 ~ #] systemctl restart httpd
[root@rhel-9 ~ #]
```

We type this in the browser: `192.168.88.160:8282`.



We can also use `curl`:

```
[root@rhel-9 ~ #] curl 192.168.88.160:8282
<h2>RHCSA is my next step</h2>
```

Task 3

Let’s download the package:

```
[root@rhel-9 ~ #] dnf install mod_ssl -y
Last metadata expiration check: 0:32:33 ago on Sun
Dependencies resolved.
=====================================================
 Package                                 Architecture
=====================================================
Installing:
 mod_ssl                                 x86_64
```

Remove the config file:

```
[root@rhel-9 ~ #] rm /etc/httpd/conf.d/ssl.conf
```

Let’s generate certificates:

```
[root@rhel-9 ~ #] openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
  -keyout /etc/pki/tls/private/mysite.key \
  -out /etc/pki/tls/certs/mysite.crt
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:az
State or Province Name (full name) []:
Locality Name (eg, city) [Default City]:
Organization Name (eg, company) [Default Company Ltd]:
Organizational Unit Name (eg, section) []:
Common Name (eg, your name or your server's hostname) []:
Email Address []:
```

Let’s create virtual host:

```
[root@rhel-9 ~ #] cd /etc/httpd/conf.d
```

```
Listen 443

<VirtualHost *:443>
    ServerName idtech.local
    DocumentRoot "/var/www/html"

    SSLEngine on
    SSLCertificateFile /etc/pki/tls/certs/mysite.crt
    SSLCertificateKeyFile /etc/pki/tls/private/mysite.key
</VirtualHost>
```

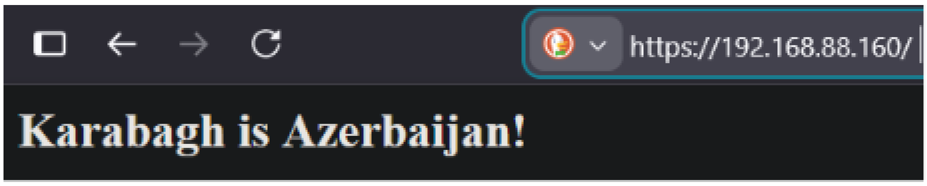
Now add the port to firewall:

```
[root@rhel-9 ~ #] firewall-cmd --add-port=443/tcp
success
[root@rhel-9 ~ #] firewall-cmd --permanent --add-port=443/tcp
success
[root@rhel-9 ~ #] firewall-cmd --reload
success
[root@rhel-9 ~ #] firewall-cmd --list-all | grep -i "ports"
  ports: 80/tcp 8282/tcp 443/tcp
```

Now restart the apache:

```
[root@rhel-9 ~ #] systemctl restart httpd
[root@rhel-9 ~ #]
```

Now we can check it in browser. We need to write just `https://192.168.88.160`. Here browser shows the certificate warning, if we proceed it:



Moreover, we can also use `curl`:

```
[root@rhel-9 ~ #] curl -k https://192.168.88.160
<h2>Karabagh is Azerbaijan!</h2>
```

Here -k flag skips certificate verification because it's self-signed.

Task 4

We need to open the virtual host file and add one line there:

```
[root@rhel-9 /etc/httpd/conf.d #] vim my-virtual-host.conf
```

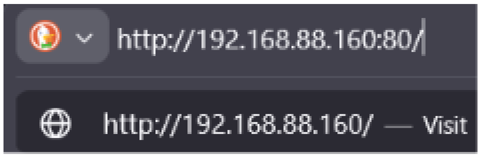
```
<VirtualHost *:80>
    ServerAdmin mr-ravan
    ServerName idtech.local
    DocumentRoot "/var/www/html"
    Redirect permanent / https://192.168.88.160/
</VirtualHost>
```

Then we need to restart apache:

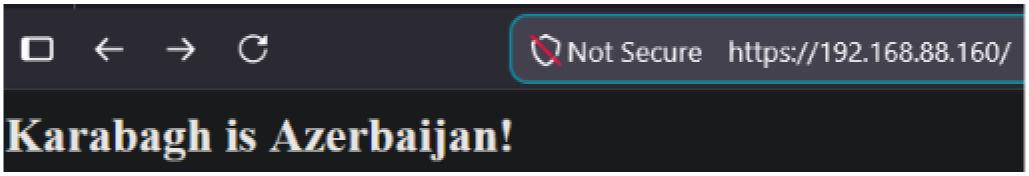
```
[root@rhel-9 /etc/httpd/conf.d #] systemctl restart httpd
[root@rhel-9 /etc/httpd/conf.d #]
```

Checking:

If we try to write `http://192.168.88.160:80/` in the browser:



(It will be redirected to https):



Task 5

Let's create the directory and index.html:

```
[root@rhel-9 ~ #] mkdir /var/www/html/protected
[root@rhel-9 ~ #] echo '<h2>Protected Content</h2>' > /var/www/html/protected/index.html
```

Let's create the password file(I gave password as `123`):

```
[root@rhel-9 ~ #] htpasswd -c /etc/httpd/.htpasswd ravan
New password:
Re-type new password:
Adding password for user ravan
[root@rhel-9 ~ #]
```

`-c` is used to create the file. If file already exists, we do not use this flag.

Let's open virtual host which we have created in task 3 and add the directory block to it.

```
[root@rhel-9 /etc/httpd/conf.d #] cat mysite-ssl.conf
Listen 443

<VirtualHost *:443>
    ServerName idtech.local
    DocumentRoot "/var/www/html"
    <Directory "/var/www/html/protected">
        AuthType Basic
        AuthName "Restricted Area"
        AuthUserFile /etc/httpd/.htpasswd
         Require valid-user
    </Directory>
    SSLEngine on
    SSLCertificateFile /etc/pki/tls/certs/mysite.crt
    SSLCertificateKeyFile /etc/pki/tls/private/mysite.key
</VirtualHost>
```

Then we should restart the service:

```
[root@rhel-9 /etc/httpd/conf.d #] systemctl restart httpd
[root@rhel-9 /etc/httpd/conf.d #]
```

Let’s check the result with `curl`:

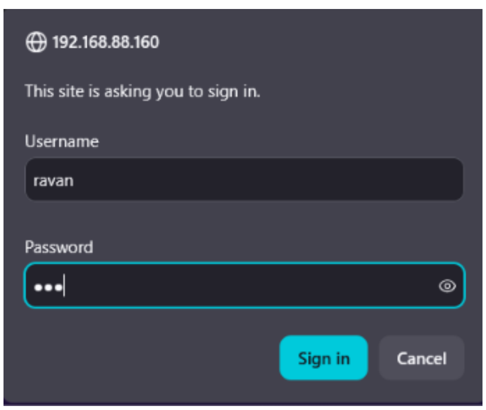
```
[root@rhel-9 /etc/httpd/conf.d #] curl -k https://192.168.88.160/protected/
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>401 Unauthorized</title>
</head><body>
<h1>Unauthorized</h1>
<p>This server could not verify that you
are authorized to access the document
requested.  Either you supplied the wrong
credentials (e.g., bad password), or your
browser doesn't understand how to supply
the credentials required.</p>
</body></html>
[root@rhel-9 /etc/httpd/conf.d #] curl -k -u ravan:123 https://192.168.88.160/protected/
<h2>Protected Content</h2>
[root@rhel-9 /etc/httpd/conf.d #]
```

In the first one there is no auth, so it does not give a result. The second one I have added auth credentials, so it gives the proper result.

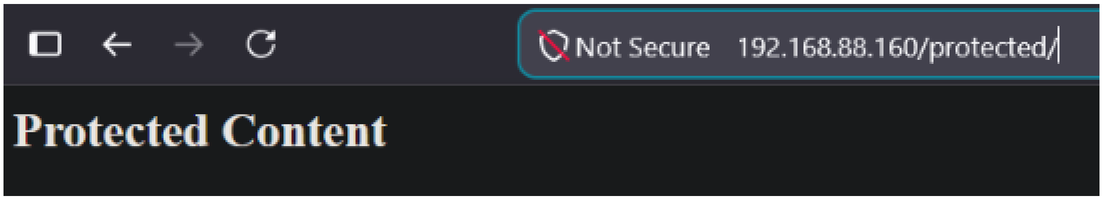
If we can look at from browser, too:

When we type it:

and press Enter, this will come:



When I enter everything correctly and click sign in, this will come:



Firewalld

Task 1

BTW, I have created `***fcmd***` as alias of `***firewall-cmd***`.

To create a new zone:

```
[root@rhel-9 ~ #] fcmd --new-zone=my-zone --permanent #Works only with permanent
success
[root@rhel-9 ~ #] fcmd --reload
success
```

We run this command to find our zone:

```
success
[root@rhel-9 ~ #] fcmd --list-all-zones
block
```

This is ours:

```
my-zone
  target: default
  icmp-block-inversion: no
  interfaces:
  sources:
  services:
  ports:
  protocols:
  forward: no
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

Task 2

I didn't understand what kind of 'service' you meant. If you meant systemd service, I have shown it below (in this task). However, if you mean firewall service, I have already handled it in Task 5 of Firewalld.

First, I create a script file inside `/usr/bin/` directory:

```
[root@rhel-9 /usr/bin #] vim my-script
[root@rhel-9 /usr/bin #] chmod +x my-script
```

```
[root@rhel-9 /usr/bin #] cat my-script
#!/bin/bash
echo "hello" > /root/FILEE
```

Let's create a service file in `/etc/system/system/` where this script above is called:

```
[root@rhel-9 /etc/systemd/system #] vim my-service.service
[root@rhel-9 /etc/systemd/system #] cat my-service.service
[Unit]
Description=my own service

[Service]
Type=oneshot
ExecStart=/usr/bin/my-script

[Install]
WantedBy=multi-user.target
```

Let's enable and start the service:

```
[root@rhel-9 /etc/systemd/system #] systemctl enable --now my-service.service
Created symlink /etc/systemd/system/multi-user.target.wants/my-service.service → /etc/systemd/system/my-service.service.
```

We can check it:

```
[root@rhel-9 /etc/systemd/system #] cat /root/FILEE
hello
```

Task 3

First, let's add this port to firewall:

```
[root@rhel-9 /usr/bin #] fcmd --add-port=8888/tcp
success
[root@rhel-9 /usr/bin #] fcmd --add-port=8888/tcp --permanent
success
[root@rhel-9 /usr/bin #] fcmd --reload
success
[root@rhel-9 /usr/bin #] fcmd --list-ports
80/tcp 443/tcp 8282/tcp 8888/tcp
[root@rhel-9 /usr/bin #]
```

Let's start listening from linux:

```
[root@rhel-9 ~ #] nc -lvp 8888
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: Listening on 0.0.0.0:8888
```

Here `-l` means listen, `-v` means verbose and `-p` means port.

Let's go to cmd in windows and type this:

```
PS C:\Users\ravan> telnet 192.168.88.160 8888
```

Afterward we can see it in linux:

```
[root@rhel-9 ~ #] nc -lvp 8888
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: Listening on 0.0.0.0:8888
Ncat: Connection from 192.168.88.1.
Ncat: Connection from 192.168.88.1:50821.
```

Task 4

If you mean to allow mysql service in firewall, let's download it:

```
[root@rhel-9 /usr/bin #] dnf whatprovides mysql
Last metadata expiration check: 0:18:48 ago on Mon 11 May 2026 02:00:05 PM +04.
mysql-8.0.43-1.el9_6.x86_64 : MySQL client programs and shared libraries
Repo        : AppStream-Repo
Matched from:
Provide     : mysql = 8.0.43-1.el9_6

[root@rhel-9 /usr/bin #] dnf install mysql-8.0.43-1.el9_6.x86_64
Last metadata expiration check: 0:18:58 ago on Mon 11 May 2026 02:00:05 PM +04.
Dependencies resolved.
==================================================================================
 Package                                                    Architecture
==================================================================================
Installing:
```

Let’s add it to firewall:

```
[root@rhel-9 /usr/bin #] fcmd --add-service=mysql
success
[root@rhel-9 /usr/bin #] fcmd --add-service=mysql --permanent
success
[root@rhel-9 /usr/bin #] fcmd --reload
success
[root@rhel-9 /usr/bin #] fcmd --list-services
cockpit dhcpv6-client mysql ntp ssh
```

(Installing the service (mysql) is not required. I did it just for fun)

Task 5

Adding it as a port is easy, we just need to run `firewall-cmd –add-port=8282/tcp –permanent` and `firewall-cmd –add-port=8282/tcp`. Now let’s do it as a service file (firewall service).

Let’s do it in a service file under firewall.

I have created a service file named `my-service.xml`:

```
[root@rhel-9 ~ #] vim /etc/firewalld/services/my-service.xml
```

Inside the file:

```
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>my-service</short>
  <description>My custom service on port 8282</description>
  <port protocol="tcp" port="8282"/>
</service>
```

Then we should reload it:

```
[root@rhel-9 ~ #] fcmd --reload
success
```

Let’s check it:

```
[root@rhel-9 ~ #] fcmd --get-services | grep "my-service"
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon ba
reos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collectd condor-c
ollector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-c
lient etcd-server finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master git gpsd
 grafana gre high-availability http http3 https ident imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin
kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-secure kube-nodeport-services
 kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp
 llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur my-service mysql nbd nebula netbios-ns netdata-dashboard nfs
 nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privox
y prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsync
d rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ssd
p ssh steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp-c
lient vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-cl
```

Now let’s add it to firewall:

```
[root@rhel-9 ~ #] fcmd --add-service=my-service
success
[root@rhel-9 ~ #] fcmd --permanent --add-service=my-service
success
[root@rhel-9 ~ #] fcmd --reload
success
[root@rhel-9 ~ #] fcmd --list-services
cockpit dhcpv6-client my-service mysql ntp ssh
```